

Erdős #1094: the least prime factor of $\binom{n}{k}$ as a domination threshold

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Abstract

#1094 asserts that for $n \geq 2k$ the least prime factor of $\binom{n}{k}$ is at most $\max(n/k, k)$, with only finitely many exceptions. We give the exact ledger form of $\text{lpf} \binom{n}{k}$ and prove that the primes above $n/2$ never contribute to it. The main result is that *every exception has $\binom{n}{k}$ squarefree*: since $\max(n/k, k) \geq \sqrt{n}$, an exception forces digit domination at every rail $p \leq \sqrt{n}$, hence every prime factor of $\binom{n}{k}$ exceeds $\sqrt{n}$ and Legendre's formula leaves each exponent at most 1. Consequently #1094 follows from finiteness of the squarefree binomial coefficients, which places it beside #175 and #378 rather than among least-prime-factor estimates. A census isolates Selfridge's $\binom{62}{6}$ as the unique exception in his conjectured range. #1094 itself is not proved.

1 Ledger form

Write $k \preceq_p n$ for base- p digit domination. By Kummer, $p \mid \binom{n}{k}$ iff the base- p addition $k + (n - k)$ carries, iff $k \not\preceq_p n$. Hence

Proposition 1. $\text{lpf} \binom{n}{k} = \min\{p : k \not\preceq_p n\}$, *computable without evaluating any binomial coefficient.*

Verified: agrees with direct computation of the least prime factor for $2 \leq k \leq 8$, $n < 400$ — no mismatch.

Lemma 2 (large rails are irrelevant). *Let $1 \leq k \leq n/2$ and $p > n/2$ prime. Then $v_p \binom{n}{k} = 1$ if $n - k < p \leq n$ and 0 if $n/2 < p \leq n - k$. In particular the least prime factor is never a prime exceeding $n/2$ unless $\binom{n}{k}$ has no smaller prime factor at all.*

Proof. For $m \leq n$ and $p > n/2$ we have $p^2 > n \geq m$, so $v_p(m!) = \lfloor m/p \rfloor$; thus $v_p \binom{n}{k} = 1 - 0 - \lfloor (n - k)/p \rfloor$ since $k \leq n/2 < p$. $\square$

So lpf is governed entirely by the small rails, in contrast to $P \binom{n}{k}$ (#683) and $\omega \binom{n}{k}$ (#685), to which the window $(n - k, n]$ contributes directly. The threshold below which $\binom{n}{k}$ is guaranteed a prime factor $\leq k$ is the subject of #1095, whose criterion $k \preceq_p n$ for all $p \leq k$ is exactly the negation of Proposition 1 restricted to $[2, k]$.

2 Every exception has a squarefree binomial coefficient

Lemma 3. *For any real $B \geq 2$, $\text{lpf} \binom{n}{k} > B$ if and only if $k \not\leq_p n$ for every prime $p \leq B$.*

Proof. By Proposition 1, $\text{lpf} \binom{n}{k} = \min\{p : k \not\leq_p n\}$, so this minimum exceeds B exactly when no prime $p \leq B$ fails domination. $\square$

Lemma 4. *For $1 \leq k \leq n$, $\max(n/k, k) \geq \sqrt{n}$.*

Proof. The two quantities have product n , so the larger is at least their geometric mean $\sqrt{n}$. $\square$

Theorem 5. *Let $n \geq 2k$ and suppose (n, k) is an exception to #1094, that is $\text{lpf} \binom{n}{k} > \max(n/k, k)$. Then every prime factor of $\binom{n}{k}$ exceeds $\sqrt{n}$, and consequently $\binom{n}{k}$ is squarefree.*

Proof. By Lemma 4 the hypothesis gives $\text{lpf} \binom{n}{k} > \sqrt{n}$, so by Lemma 3 we have $k \not\leq_p n$ for every prime $p \leq \sqrt{n}$; equivalently no prime $p \leq \sqrt{n}$ divides $\binom{n}{k}$. Hence every prime factor p of $\binom{n}{k}$ satisfies $p > \sqrt{n}$, so $p^2 > n$ and Legendre's formula has a single term: $v_p \binom{n}{k} = \lfloor n/p \rfloor - \lfloor k/p \rfloor - \lfloor (n-k)/p \rfloor \leq 1$. Every exponent in the factorisation of $\binom{n}{k}$ is therefore at most 1. $\square$

Verified: all 14 exceptions with $n < 4000$, $k < 30$ — namely $(7, 3), (13, 4), (14, 4), (23, 5), (62, 6), (44, 8), (46, 10)$ — have $\binom{n}{k}$ squarefree and $\text{lpf} > \sqrt{n}$, with no exception to either step.

Corollary 6. *#1094 follows from the statement that $\binom{n}{k}$ is squarefree for only finitely many pairs (n, k) with $n \geq 2k$ and $k \geq 2$.*

Proof. Immediate from Theorem 5: the exception set injects into the set of pairs with $\binom{n}{k}$ squarefree. $\square$

This places #1094 in the same circle as Erdős #175 (that $\binom{2n}{n}$ is never squarefree for $n \geq 5$, proved by Sárközy and by Granville–Ramaré) and #378 (the density of n with $\binom{n}{k}$ squarefree, resolved from Granville–Ramaré), rather than in the circle of least-prime-factor estimates where it is usually placed.

3 Census

Exceptions to $\text{lpf} \binom{n}{k} \leq \max(n/k, k)$ over $n < 3000$, $k < 25$: **13** in total, beginning

$$(n, k, \text{lpf}) = (7, 3, 5), (13, 4, 5), (14, 4, 7), (23, 5, 7), (62, 6, 19), (44, 8, 11).$$

Restricting to Selfridge's range $n \geq k^2 - 1$ leaves **exactly one**: $(62, 6, 19)$, i.e. $\binom{62}{6}$ — precisely and only the exception he named. The ledger form therefore reproduces the unique counterexample of a published conjecture without being given it.

Remark 7 (register). **Proved:** Proposition 1 and Lemma 2, elementary from Kummer and Legendre; neither is claimed new. Lemmas 3, 4, Theorem 5 and Corollary 6: every exception to #1094 has $\binom{n}{k}$ squarefree, so #1094 follows from finiteness of the squarefree binomial coefficients. **Measured:** the 13 exceptions and the isolation of $\binom{62}{6}$ in the range $n \geq k^2 - 1$. **Not proved:** finiteness of the exception set, i.e. #1094 itself. **Falsifier:** a second exception with $n \geq k^2 - 1$ would refute Selfridge's conjecture; the domination form is cheap enough to search $k \leq 100$, $n \leq 10^6$ routinely, and such a hit would be worth reporting on its own.